TabPFN on Tabular Data

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1 Introduction

We evaluate **TabPFN v2.6** [1] on 10 tabular datasets (5 classification, 5 regression) from COMP 4332 Project 2, comparing in-context learning and fine-tuning against the provided baselines. Results show that TabPFN consistently surpasses all baselines across every dataset.

2 Model and Configuration

2.1 Motivation of Model Choice

We adopt **TabPFN v2.6** [1] for all ten datasets. TabPFN is a Prior-Data Fitted Network that performs supervised learning via in-context learning over a transformer, predicting test labels in a single forward pass without gradient updates [2]. We select version 2.6 specifically because it shares the advanced architecture of TabPFN-2.5 while further pushing state-of-the-art performance on tabular benchmarks, yielding demonstrably superior accuracy and regression performance over previous versions [1]. Since the `tabpfn` library's fine-tuning API defaults to v2.5 with no configuration option, we patched the source code to enforce v2.6.

2.2 Hardware Setting

Prediction and training are conducted on a workstation with an Intel Core i9-13900K CPU and an NVIDIA GeForce RTX 4090 D GPU (48 GB VRAM).

2.3 Choice of Context

TabPFN's context window imposes a hard limit on training rows (e.g. `cls/3` has 46,602 rows). We implement two task-specific subsampling strategies.

- **Classification — Stratified Sampling.** We enforce a minimum quota per class (30 samples) to prevent minority classes from being dropped, then distribute the remaining budget proportionally to class frequency. When a class has fewer samples than the quota, we oversample with replacement.
- **Regression — Quantile-Bucket Sampling.** We discretise the continuous target into 20 equal-frequency bins and allocate a minimum of 100 samples per bin, with the remainder distributed proportionally. This ensures the context covers the full range of the target distribution, including the tails, which is critical for RMSE.

We sweep maximum context sizes $N \in \{5000, 7500, 10000, 20000, 30000, 40000\}$ and evaluate both raw features and `StandardScaler`-normalised float features. For fine-tuning, under limitation of VRAM, we automatically selects the maximized feasible context size from $\{10000, 5000, 4000, 3000, 2000, 1000\}$ independently for each dataset.

3 Comparison of Model Performances

3.1 Effect of Context Size and Feature Scaling

Figure 1 shows how performance varies with context size, normalised as a percentage improvement relative to the 5k context baseline. For classification, accuracy improves monotonically with context size on most datasets; `cls/1` gains the most (+7.2% relative improvement from 5k to 40k). For regression, `reg/2` benefits substantially from larger context (RMSE drops by over 25% relative to 5k), while `reg/3` and `reg/4` saturate early. Feature scaling (solid lines) shows *no consistent benefit* over raw features (dashed lines): on several datasets it slightly hurts performance, confirming that TabPFN's internal normalisation is already effective.

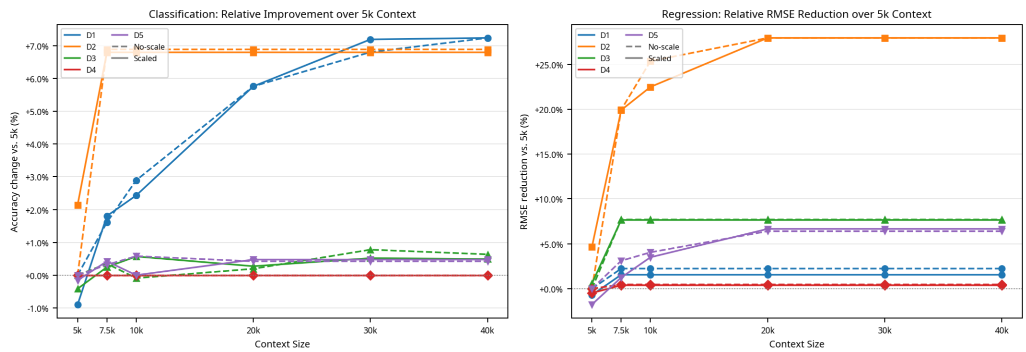

Figure 1: Relative performance improvement vs. context size, using 5k context (no-scale) as the 0% reference point. Left: Classification accuracy gain. Right: Regression RMSE reduction. Solid lines = scaled features; dashed lines = raw features.

3.2 Final Results: Baseline vs. Context-only (40k) vs. Fine-tuned

Table 1 and Figure 2 summarise the final comparison. The context-only model (40k, no scaling) already outperforms the baseline on *all* 10 datasets. Fine-tuning provides additional gains on 9 out of 10 datasets, with the largest improvements on `cls/2` (+19.2% relative to baseline) and `reg/2` (34.3% RMSE reduction relative to baseline). The submitted `predict.csv` files are generated from the fine-tuned models.

Table 1: Test-set performance. Bold = best among the three methods. ↑ higher is better; ↓ lower is better.

Classification (Accuracy ↑)						Regression (RMSE ↓)					
Method	**D1**	**D2**	**D3**	**D4**	**D5**	**Method**	**D1**	**D2**	**D3**	**D4**	**D5**
Baseline	0.860	0.700	0.640	0.680	0.700	Baseline	2.600	3.600	0.700	0.030	0.210
Context (40k)	0.934	0.785	0.667	0.707	0.717	Context (40k)	1.950	2.686	0.582	0.026	0.195
Fine-tuned	**0.941**	**0.834**	0.666	**0.709**	**0.717**	Fine-tuned	**1.945**	**2.346**	**0.580**	**0.026**	**0.191**

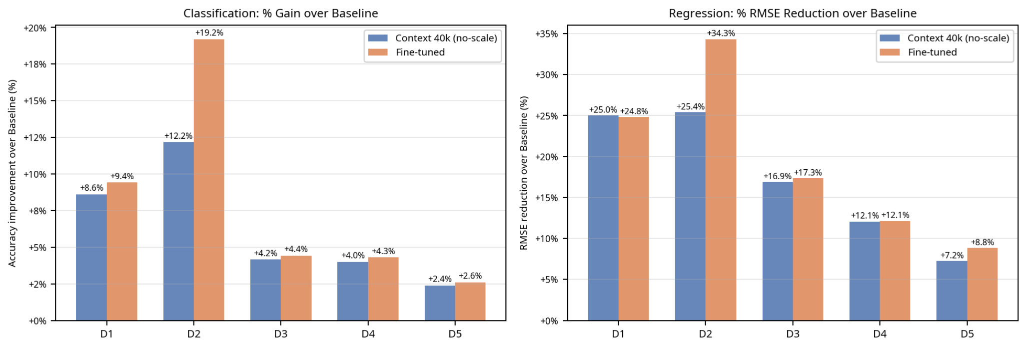

Figure 2: Relative improvement over the provided baseline (%). Left: Accuracy gain. Right: RMSE reduction.

4 Conclusion

TabPFN v2.6 with stratified/quantile-bucket context sampling surpasses the provided baseline on all 10 datasets without any gradient updates. Larger context sizes consistently improve performance, and explicit feature scaling is unnecessary. Fine-tuning further boosts accuracy and reduces RMSE on nearly all datasets, with the most notable gains on datasets that have larger or more complex training distributions (`cls/2`, `reg/2`). The combination of in-context inference and lightweight fine-tuning makes TabPFN a highly practical solution for tabular prediction tasks.

References

[1] Léo Grinsztajn, Klemens Flöge, Oscar Key, Felix Birkel, Brendan Roof, Phil Jund, Benjamin Jäger, Adrian Hayler, Dominik Safaric, Simone Alessi, Felix Jablonski, Mihir Manium, Rosen Yu, Anurag Garg, Jake Robertson, Shi Bin Hoo, Vladyslav Moroshan, Magnus Bühler, Lennart Purucker, Clara Cornu, Lilly Charlotte Wehrhahn, Alessandro Bonetto, Sauraj Gambhir, Noah Hollmann, and Frank Hutter. TabPFN-2.5: Advancing the state of the art in tabular foundation models, 2025. URL https://arxiv.org/abs/2511.08667. TabPFN-2.6 shares the same architecture and dataset constraints as TabPFN-2.5, further pushing state-of-the-art performance.

[2] Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations (ICLR)*, 2023. URL https://openreview.net/forum?id=cp5PvcI6w8_.